Spinning Machine



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Spinning Machine

What it is used for

A spinning machine twists fibres together to form yarn of the required thickness and strength. It may be operated manually or by motor, depending on the design and textile process.

Main parts

- Identify the fibre feeding area and drafting or fibre preparation rollers.
- Locate the spindle, rotor or twisting mechanism used to form yarn.
- Identify the bobbin, spool or yarn collection arrangement.
- Locate the drive belt, motor and transmission components, where fitted.
- Identify the speed control and start-stop switches.
- Locate the thread guides, tension controls and protective guards.
- Identify the emergency stop and cleaning access points.

Before you start

1. Read the operating instructions and obtain teacher supervision before use.
2. Inspect the spindle, rollers, belts, thread guides and guards for damage.
3. Ensure fibres are clean and suitable for the machine type.
4. Tie back long hair and remove jewellery and loose clothing.
5. Keep hands clear of rotating components and avoid loose gloves.
6. Check that bobbins are correctly fitted and the work area is uncluttered.
7. Confirm that electrical connections and emergency controls are functioning properly.

How to use

1. Switch off the machine before threading fibres or fitting a bobbin.
2. Position the prepared fibres in the designated feeding arrangement.
3. Guide the fibre through the specified rollers and thread guides.
4. Attach the fibre end securely to the empty bobbin or collection point.
5. Set the recommended speed and tension for the fibre and yarn type.
6. Close guards and confirm that loose tools are removed.
7. Start the machine and observe the initial yarn formation.
8. Feed fibres steadily and monitor yarn thickness and thread tension.
9. Stop the machine before correcting broken yarn or adjusting the thread path.
10. Switch off and wait for rotating parts to stop before removing full bobbins.

⚠ Safety

- Never touch rotating spindles, rollers or drive belts.
- Do not wear loose gloves, jewellery or loose sleeves near moving parts.
- Switch off and isolate power before clearing tangled fibres or servicing.
- Keep guards fitted and never bypass safety switches.
- Avoid inhaling accumulated fibre dust and maintain suitable ventilation.
- Report unusual vibration, noise or overheating to the teacher.

Care & storage

- Remove fibre lint and dust after switching off and isolating power.
- Clean thread guides and rollers using approved tools.
- Inspect belts, bearings, spindles and guards for wear or damage.
- Store bobbins, tools and the machine in a clean, dry area.

Common mistakes to avoid

- Feeding fibres unevenly and producing inconsistent yarn thickness.
- Operating with tangled fibres or incorrectly fitted bobbins.
- Reaching into rotating parts to repair broken yarn.
- Ignoring worn belts, damaged guards or excessive fibre accumulation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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